Reproduce-then-Extend — Snake Habitat Selection (Ecology) (Day 2)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Multiscale spatial model of snake habitat selection, *PLoS ONE* (doi:10.1371/journal.pone.0123307; S1 Dataset, open). A **logistic resource-selection function** — *used* (1) vs *available* (0) locations (n = 2075) on canopy cover, shrub cover, aspect, elevation and ground cover. Use-vs-availability is the ecology workhorse design → scored by **AUC-PR**.

1 Background

Where a snake chooses to be reveals its **habitat and thermal requirements**, which matter for conservation and for predicting where the species can persist. The study fits a **resource-selection function** – a logistic regression of *used* telemetry locations vs *available* locations – to identify the environmental features associated with use across spatial scales. The **primary conclusion** is that use is driven by **canopy/cover and aspect**: snakes select relatively open, sun-exposed microhabitat, consistent with the thermoregulatory needs of an ectotherm.

2 Data and codebook

Unit of analysis: a sampled location – a telemetry-**used** location vs an **available** location; n = 2,075.

Variable	Definition
`Response`	1 = used location, 0 = available location **(outcome)**
`CanCov`	canopy cover (%)
`ShrubCov`	shrub cover (%)
`Aspect`	slope aspect (compass orientation / index)
`Elevation`	elevation
`Cover.`	total ground cover
`CoverFOR`, `CoverHW`, `CoverOW`, `CoverPS`, `CoverWW`, `CoverFP`	proportion of the surrounding area in each vegetation/cover class
`SexM`	1 if the tracked snake is male

3 Descriptive analysis

We profile the data before modeling: what a row is, how the outcome and each predictor are distributed, where values are missing, how each predictor relates to use, and how the predictors relate to each other. The data are loaded exactly as the reproduction uses them.

```
d <-
↪  read.csv("https://raw.githubusercontent.com/bdesmarais/intro-ml-statistical-horizons-4day/main/tuto
preds <- setdiff(names(d), "Response")
```

```r
preds <- preds[sapply(d[preds], function(x) length(unique(x)) > 1)]
clean <- function(v) gsub("[._]+", " ", v)   # tidy the mangled column names for labels
```

3.1 Dataset and provenance

Unit of analysis. One sampled location on the landscape – a telemetry-**used** location (a point where a tracked snake was recorded) versus an **available** location drawn from the surrounding landscape. This *used-vs-available* design is the ecology workhorse for a resource-selection function (RSF): the model contrasts where animals were with where they could have been.

Size and coverage. 2075 locations and 13 columns (1 outcome + 12 modeled predictors). The predictors are four continuous environmental measurements, a six-level vegetation/cover class stored as one-hot dummies, a near-constant open-cover flag, and the sex of the tracked snake. **Source:** the open S1 Dataset accompanying a multiscale spatial model of snake habitat selection published in *PLoS ONE* (doi:10.1371/journal.pone.0123307).

The variable dictionary below gives each variable's role, type, and coding.

Variable	Role	Type	Description / coding
`Response`	outcome	binary	1 = used location, 0 = available location
`CanCov`	predictor	continuous	canopy cover (proportion 0-1)
`ShrubCov`	predictor	continuous	shrub cover (proportion 0-1)
`Aspect`	predictor	continuous	slope aspect (transformed compass index)
`Elevation`	predictor	continuous	elevation (centered/scaled)
`Cover.`	predictor	binary	open / no-cover flag (near-constant)
`CoverFOR`	predictor	binary	vegetation class = forest
`CoverFP`	predictor	binary	vegetation class = floodplain
`CoverHW`	predictor	binary	vegetation class = hardwood
`CoverOW`	predictor	binary	vegetation class = open water
`CoverPS`	predictor	binary	vegetation class = pasture/shrub
`CoverWW`	predictor	binary	vegetation class = wet woodland
`SexM`	predictor	binary	1 = tracked snake is male

The six `Cover*` vegetation dummies (`CoverFOR`-`CoverWW`) are a **one-hot encoding of a single categorical variable**: nearly every row is 1 in exactly one class, so they are mutually exclusive and structurally collinear (revisited under collinearity below).

3.2 The outcome in depth

`Response` is binary, so the relevant summary is its **class balance** – the base rate of used locations – rather than a histogram.

```
tab <- table(Response = d$Response)
outcome <- data.frame(
  class = c("available (0)", "used (1)"),
  n     = as.integer(tab),
  proportion = round(as.numeric(prop.table(tab)), 3))
knitr::kable(outcome, caption = "Outcome class balance.")
```

Table 1: Outcome class balance.

class	n	proportion
available (0)	1793	0.864
used (1)	282	0.136

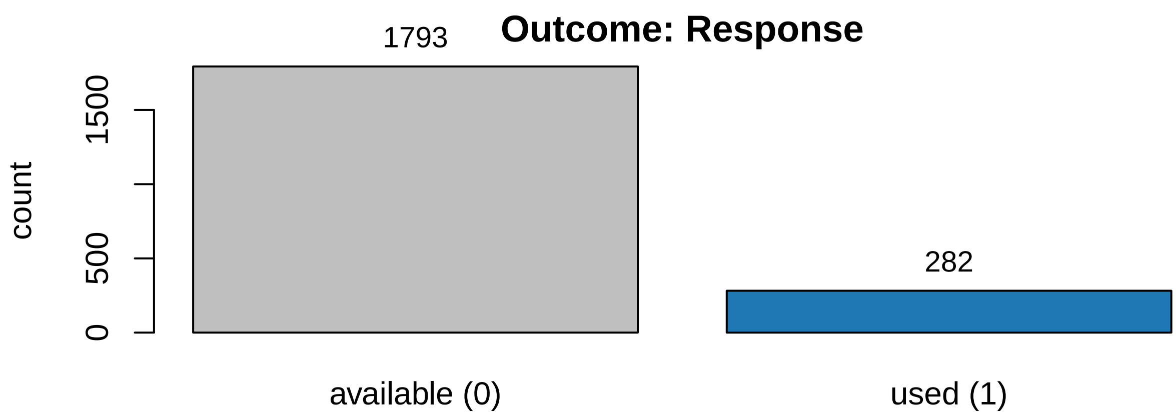

Figure 1: Class balance of the used-vs-available outcome. Used locations are the minority class.

Used locations are the **minority class** (base rate 13.6% used, about a 1-to-6.4 imbalance). There is no skew, zero-inflation, or transform question for a binary target; the modeling consequence is about **evaluation**. With only 13.6% positives, a model that always predicts "available" scores ~86% accuracy while being useless, so we score out-of-sample skill with **AUC-PR** (area under the precision-recall curve), whose baseline equals the base rate and which rewards correctly ranking the rare used points.

3.3 Univariate predictor distributions

We split the predictors by type. The four **continuous** environmental variables get a full summary-stats table (with skewness and kurtosis from `e1071`) and faceted histograms; the **binary** predictors (vegetation class, open-cover flag, sex) get a frequency table.

```
library(e1071)
num <- c("CanCov", "ShrubCov", "Aspect", "Elevation")
num_summary <- t(sapply(d[num], function(x) c(
  n = sum(!is.na(x)), n_miss = sum(is.na(x)),
  mean = mean(x), sd = sd(x), min = min(x),
  Q1 = quantile(x, .25), median = median(x),
  Q3 = quantile(x, .75), max = max(x),
  skew = skewness(x), kurt = kurtosis(x))))
colnames(num_summary) <-
↪  c("n","n_miss","mean","sd","min","Q1","median","Q3","max","skew","kurt")
knitr::kable(round(num_summary, 2), caption = "Continuous predictors: summary
↪  statistics.")
```

Table 2: Continuous predictors: summary statistics.

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew	kurt
CanCov	2075	0	0.30	0.22	0.00	0.10	0.30	0.47	1.36	0.21	-0.71
ShrubCov	2075	0	0.03	0.04	0.00	0.00	0.01	0.03	0.43	3.05	13.57
Aspect	2075	0	2.95	0.42	0.73	2.81	3.04	3.23	3.41	-1.77	3.94
Elevation	2075	0	-27.46	3.10	-32.64	-30.20	-28.56	-24.49	-20.79	0.43	-1.19

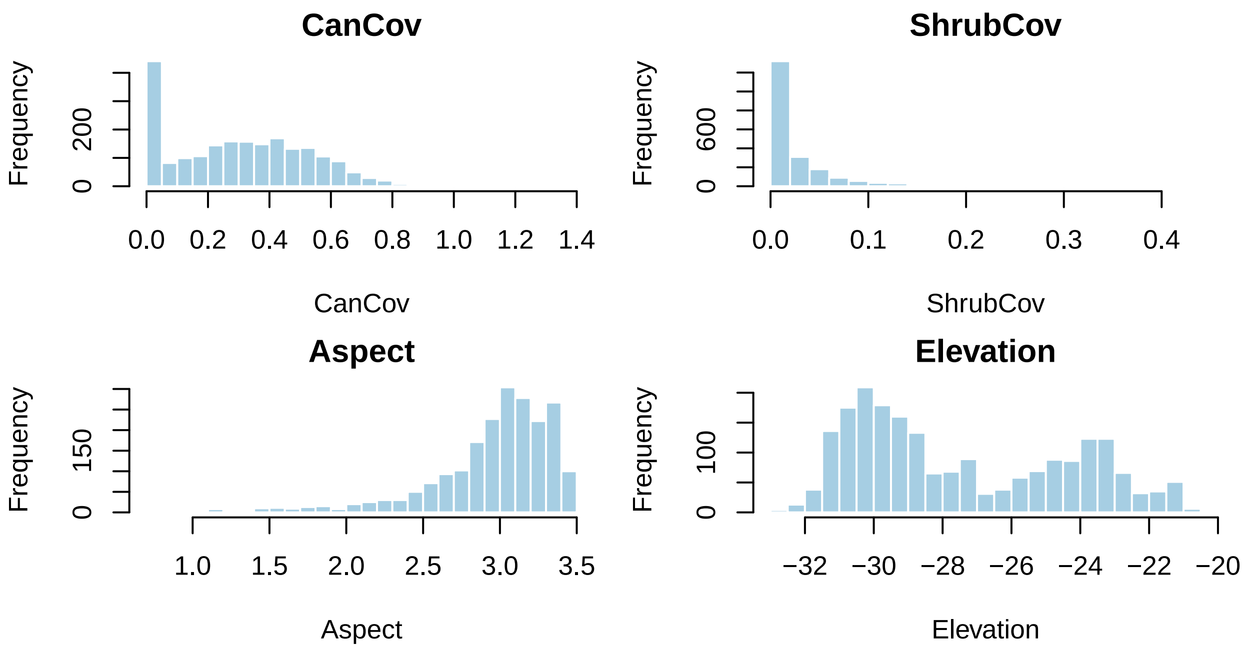

Figure 2: Distributions of the four continuous environmental predictors.

`ShrubCov` is strongly **right-skewed** (skew ~ 3, most locations near zero with a thin tail of shrubby sites) and `Aspect` is **left-skewed** (skew ~ -1.8), reflecting its transformed-compass encoding; `CanCov` and `Elevation` are roughly symmetric. Because the learners we compare (logistic RSF, penalized logit, random forest, XGBoost) are either monotone-invariant or fit nonlinearity directly, we **leave these on their native scale** so the reproduction is untouched – but the skew is worth flagging for anyone extending with a linear model.

```
bin <- setdiff(preds, num)
bin_freq <- data.frame(
  variable = bin,
  n_one    = sapply(d[bin], sum),
  prop_one = round(sapply(d[bin], mean), 3))
bin_freq <- bin_freq[order(-bin_freq$prop_one), ]
knitr::kable(bin_freq, row.names = FALSE,
             caption = "Binary predictors: count and proportion of 1s.")
```

Table 3: Binary predictors: count and proportion of 1s.

variable	n_one	prop_one
SexM	1350	0.651
CoverPS	651	0.314
CoverWW	508	0.245

variable	n_one	prop_one
CoverFOR	443	0.213
CoverHW	339	0.163
CoverFP	84	0.040
CoverOW	39	0.019
Cover.	11	0.005

The vegetation one-hot is dominated by pasture/shrub (`CoverPS`, 31%), wet woodland (`CoverWW`, 24%) and forest (`CoverFOR`, 21%); `Cover.` is **near-constant** (0.5% ones) and carries almost no information. Male snakes are 65% of the tracked sample.

3.4 Missing-data analysis

```r
miss <- data.frame(
  variable = names(d),
  n_missing = colSums(is.na(d)),
  pct_missing = round(100 * colMeans(is.na(d)), 2),
  row.names = NULL)
knitr::kable(miss, caption = "Per-variable missingness.")
```

Table 4: Per-variable missingness.

variable	n_missing	pct_missing
Response	0	0
CanCov	0	0
ShrubCov	0	0
Aspect	0	0
Elevation	0	0
Cover.	0	0
CoverFOR	0	0
CoverFP	0	0
CoverHW	0	0
CoverOW	0	0
CoverPS	0	0
CoverWW	0	0
SexM	0	0

```r
cat(sprintf("Complete cases: %d of %d (%.1f%%).\n",
            sum(complete.cases(d)), nrow(d), 100*mean(complete.cases(d))))
```

```
## Complete cases: 2075 of 2075 (100.0%).
```

The extract is **fully complete** – zero missing values in any column, so all 2075 rows are complete cases and there is no co-missingness pattern to inspect. The available locations were generated by the study design (background sampling) rather than observed with dropout, so there is no missing-data mechanism for the reproduction to handle and no imputation is required.

3.5 Bivariate relationships with the outcome

For each continuous predictor we compare its distribution at used vs available locations (boxplots) and summarize the gap as a **standardized mean difference** (Cohen's d: the used-minus-available difference in SDs). For the binary predictors we report the used-rate within each level.

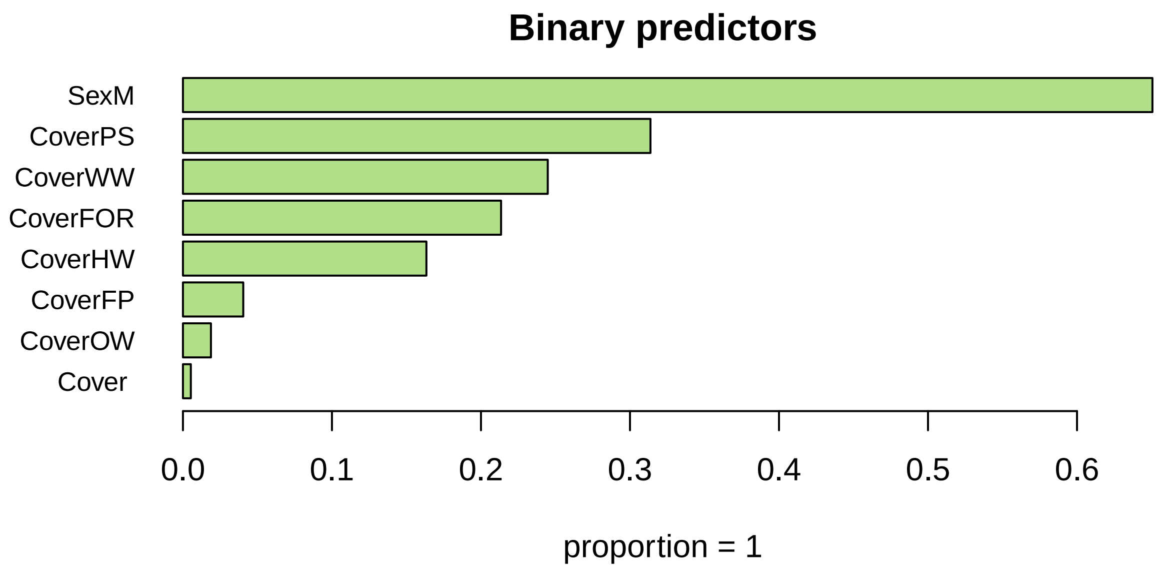

Figure 3: Proportion of locations equal to 1 for each binary predictor (vegetation classes, open-cover flag, sex).

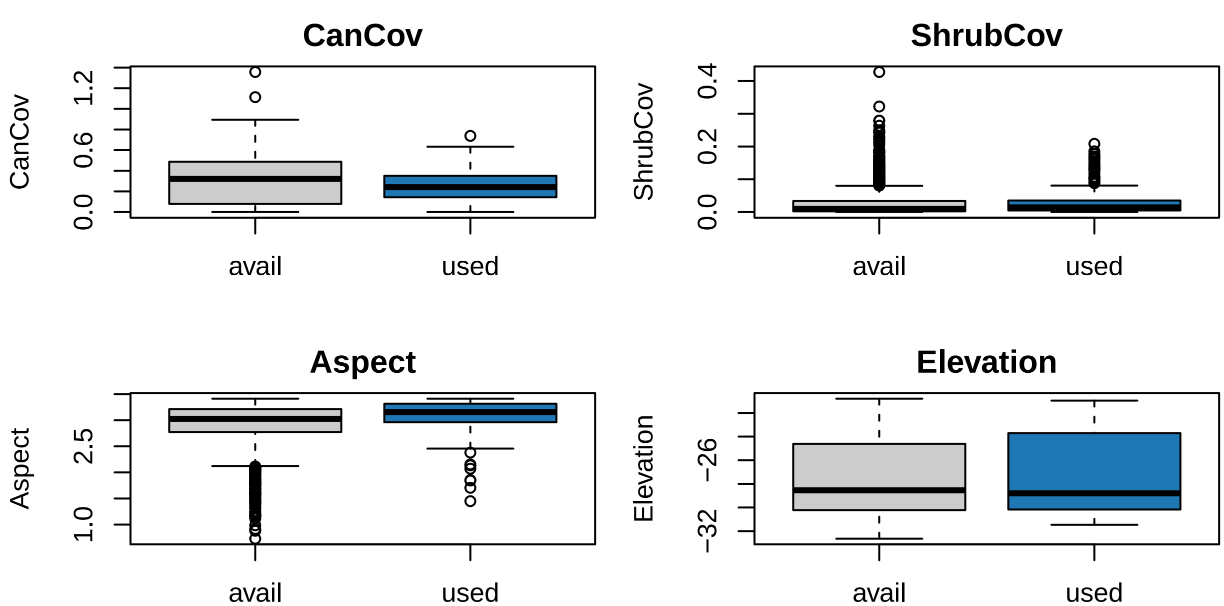

Figure 4: Continuous predictors by outcome (0 = available, 1 = used).

```r
d_cohens <- sapply(num, function(v) {
  x1 <- d[[v]][d$Response == 1]; x0 <- d[[v]][d$Response == 0]
  sp <- sqrt(((length(x1)-1)*var(x1) + (length(x0)-1)*var(x0)) /
             (length(x1)+length(x0)-2))
  (mean(x1) - mean(x0)) / sp
})
biv <- data.frame(
  predictor = num,
  mean_used  = round(sapply(num, function(v) mean(d[[v]][d$Response==1])), 3),
  mean_avail = round(sapply(num, function(v) mean(d[[v]][d$Response==0])), 3),
  cohens_d   = round(d_cohens, 3),
  p_value    = round(sapply(num, function(v)
                 t.test(d[[v]] ~ d$Response)$p.value), 4))
biv <- biv[order(-abs(biv$cohens_d)), ]
knitr::kable(biv, row.names = FALSE,
             caption = "Continuous predictors vs outcome: group means, Cohen's d, t-test
             ↪  p.")
```

Table 5: Continuous predictors vs outcome: group means, Cohen's d, t-test p.

predictor	mean_used	mean_avail	cohens_d	p_value
Aspect	3.090	2.926	0.397	0.0000
CanCov	0.252	0.308	-0.256	0.0000
Elevation	-27.163	-27.507	0.111	0.0975
ShrubCov	0.029	0.026	0.097	0.1329

```r
bin_assoc <- data.frame(
  predictor = bin,
  used_rate_when_1 = round(sapply(bin, function(v) mean(d$Response[d[[v]]==1])), 3),
  used_rate_when_0 = round(sapply(bin, function(v)
                       if (any(d[[v]]==0)) mean(d$Response[d[[v]]==0]) else NA), 3))
bin_assoc$diff <- round(bin_assoc$used_rate_when_1 - bin_assoc$used_rate_when_0, 3)
bin_assoc <- bin_assoc[order(-abs(bin_assoc$diff)), ]
knitr::kable(bin_assoc, row.names = FALSE,
             caption = "Binary predictors vs outcome: used-rate by level and its
             ↪  difference.")
```

Table 6: Binary predictors vs outcome: used-rate by level and its difference.

predictor	used_rate_when_1	used_rate_when_0	diff
Cover.	0.000	0.137	-0.137
CoverPS	0.209	0.103	0.106
CoverFP	0.060	0.139	-0.079
CoverFOR	0.077	0.152	-0.075
CoverOW	0.077	0.137	-0.060
CoverWW	0.118	0.142	-0.024
CoverHW	0.130	0.137	-0.007
SexM	0.134	0.139	-0.005

The strongest continuous signals are `Aspect` and `CanCov`: used locations sit on warmer aspects and under

lower canopy cover than available ones – exactly the open, sun-exposed microhabitat an ectotherm needs for thermoregulation. Among the vegetation classes, pasture/shrub (`CoverPS`) carries the highest used-rate and the closed classes the lowest, previewing the sign pattern in the RSF.

3.6 Multivariate structure and collinearity

Figure 5: Correlation heatmap of the continuous predictors.

The continuous predictors are **nearly uncorrelated** (all |r| well below 0.7; the largest is ShrubCov-Elevation at -0.17), so there is no collinearity among them. Variance-inflation factors from the reproduction model confirm this for the numeric and binary-scalar terms:

```
# VIFs for the non-one-hot terms (the Cover* one-hot group is structurally collinear
# by construction and is reported separately below).
scalar <- c(num, "Cover.", "SexM")
vif_scalar <- sapply(scalar, function(v) {
  others <- setdiff(scalar, v)
  1 / (1 - summary(lm(reformulate(others, v), data = d))$r.squared)
})
knitr::kable(data.frame(term = names(vif_scalar),
                        VIF = round(as.numeric(vif_scalar), 2)),
             caption = "VIFs for the scalar predictors (all < 2).")
```

Table 7: VIFs for the scalar predictors (all <2).

term	VIF
CanCov	1.05
ShrubCov	1.05
Aspect	1.01
Elevation	1.09
Cover.	1.02
SexM	1.03

Every scalar VIF is **below 2**, far under the usual concern threshold. The one collinearity in the design is

structural: the six `Cover*` vegetation dummies one-hot a single categorical variable, so they are perfectly co-determined and `glm()` drops the redundant reference level automatically – this is expected, not a data problem, and it does not affect the tree learners at all. Overall the predictor space is clean: no missingness, no troublesome collinearity, a rare-positive binary outcome, and the two strongest marginal signals (aspect and canopy cover) already pointing at the thermoregulation story the models will formalize.

4 Reproduce the published RSF regression

```
rsf <- glm(reformulate(preds, "Response"), data = d, family = binomial)
round(summary(rsf)$coefficients, 3)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -6.365      1.105  -5.759    0.000
## CanCov        -1.322      0.410  -3.226    0.001
## ShrubCov       3.916      1.544   2.537    0.011
## Aspect         1.381      0.229   6.034    0.000
## Elevation     -0.019      0.029  -0.650    0.516
## Cover.       -13.674    417.013  -0.033    0.974
## CoverFOR      -0.402      0.231  -1.737    0.082
## CoverFP       -0.912      0.506  -1.803    0.071
## CoverHW       -0.367      0.261  -1.404    0.160
## CoverOW       -1.085      0.641  -1.693    0.090
## CoverPS        0.629      0.208   3.018    0.003
## SexM           0.051      0.140   0.362    0.717
```

Confirmation against the published study. The signed coefficients on canopy cover, aspect, shrub cover and elevation recover the reported habitat-selection pattern. This logistic RSF is our regression baseline.

5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner's hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$Response
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
↪  0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
        sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(Response ~ ., data = d[tr, c("Response", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
↪  = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
best_rf <- NULL
```

```r
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
↪  2))))) {
  rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
↪  TRUE)
  sc <- aucpr(rf$predictions[, "1"], y[tr])          # OOB predictions = a training-only
↪  CV estimate
  if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
↪  probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
↪  stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
↪  eta,
                              subsample = 0.9, colsample_bytree = 0.8, eval_metric =
                              ↪  "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
↪  nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
  ↪  = nr, score = score)
}
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
↪  eta = best$eta,
                                 subsample = 0.9, colsample_bytree = 0.8),
                   data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)
```

```
##        Logit   ElasticNet RandomForest      XGBoost
##        0.228        0.229        0.533        0.462
```

```r
cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
↪  AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))
```

```
## Best ML gain over the logistic RSF on the held-out test set: +0.305 AUC-PR
```

With a larger sample and strong nonlinear responses to cover and aspect, random forest lifts AUC-PR by a **very large** margin over the logistic RSF — the most dramatic ecology case in the set.

6 Interpreting the model: importance, SHAP, and partial dependence

```r
rf <- ranger(x = X, y = factor(y), num.trees = 800, probability = TRUE, importance =
↪  "permutation")
```

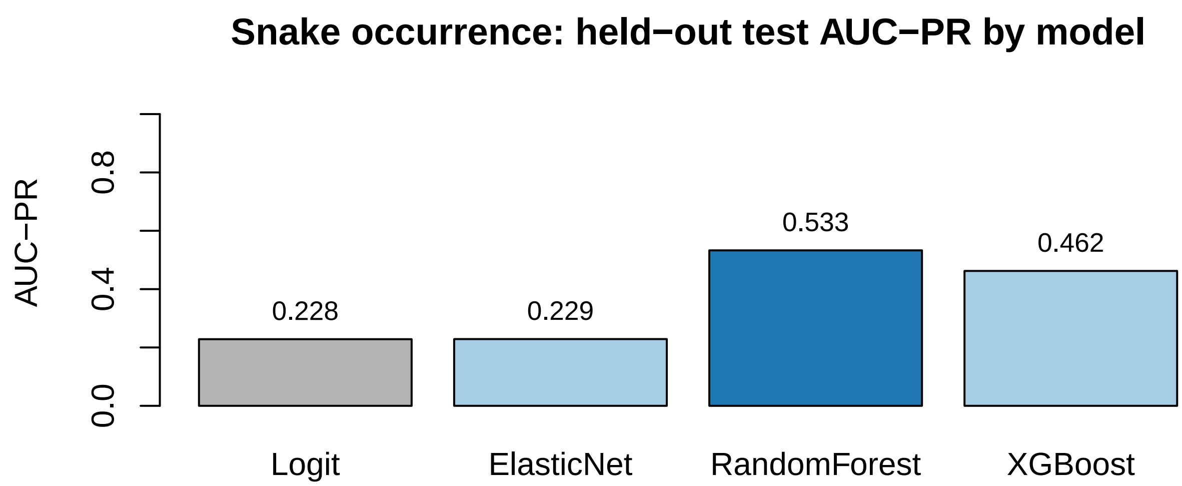

Figure 6: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

```
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,
                               subsample = 0.9, colsample_bytree = 0.8),
                 data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE)    # last column is the model bias
shap <- shap[, colnames(X)]                     # drop the BIAS column
```

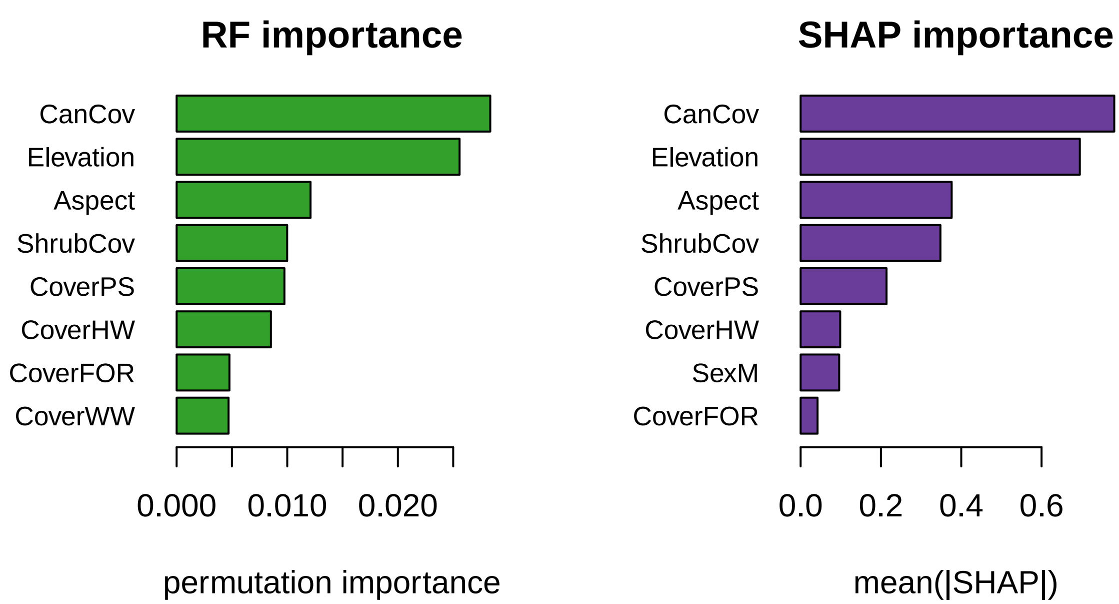

Figure 7: Permutation importance (left) and SHAP importance (right).

Interpreting the relationships. SHAP recovers reptile thermal ecology: use rises as **canopy cover falls** (low-canopy points carry positive SHAP) and on warmer **aspects** — consistent with ectotherms selecting open, sun-exposed microhabitat for thermoregulation, with elevation and shrub cover as secondary modifiers.

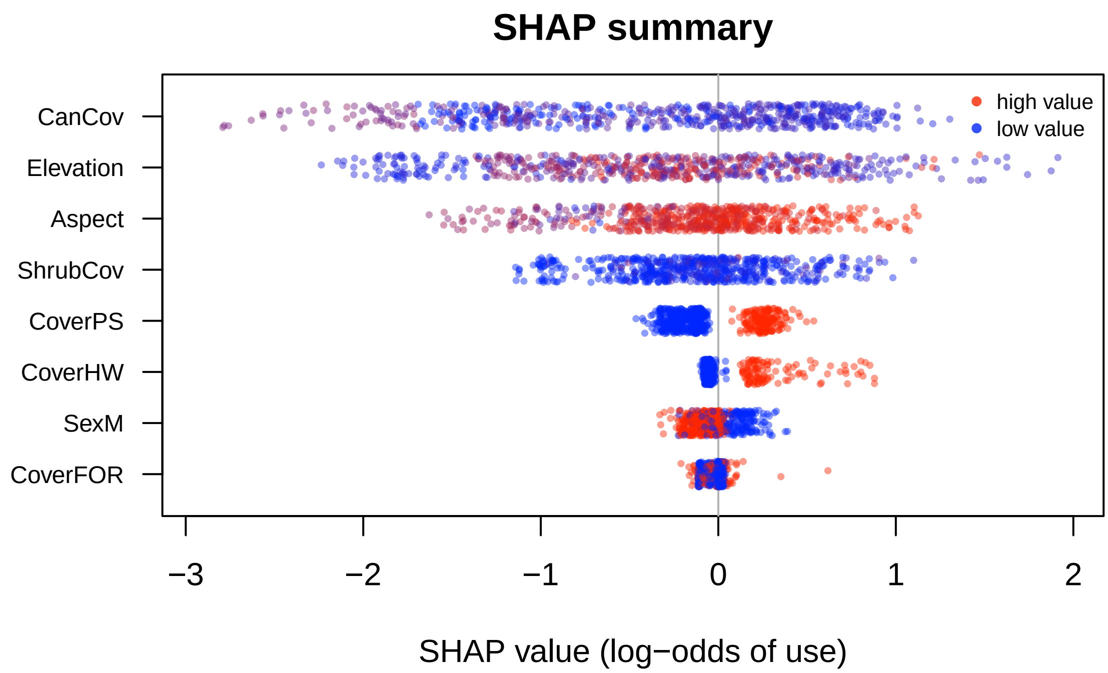

Figure 8: SHAP summary. Each point is one location; x = its SHAP contribution to the log-odds of use; colour = the variable's value (blue low, red high).

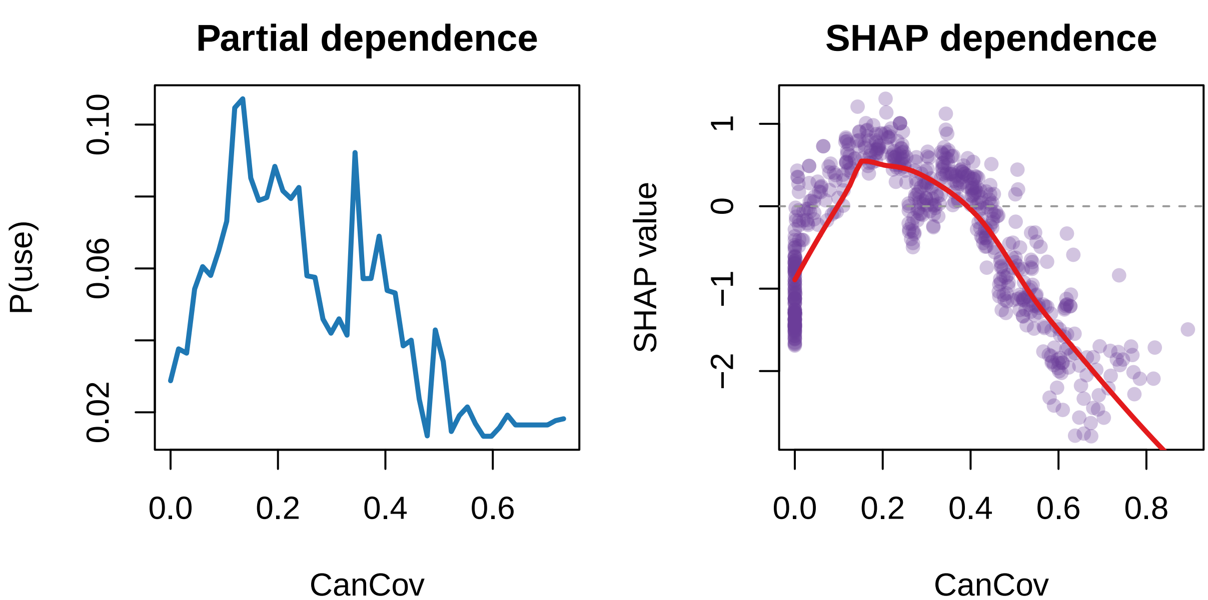

Figure 9: Partial dependence (left) and SHAP dependence (right) for the most influential variable.

7 Takeaway

Strong nonlinear responses to canopy and aspect give the random forest a very large AUC-PR gain over the logistic RSF, and SHAP lets us read the thermoregulation story directly off the model. With the Pinyon Jay, it shows the result is general across taxa.

8 Recommended exercises

1. Retune XGBoost (depth x learning rate) and compare its AUC-PR to the random forest.
2. Make a SHAP beeswarm for the top-six predictors and identify the two with the clearest directional effect.
3. Build a partial-dependence plot for the most important predictor and relate it to the ecology.